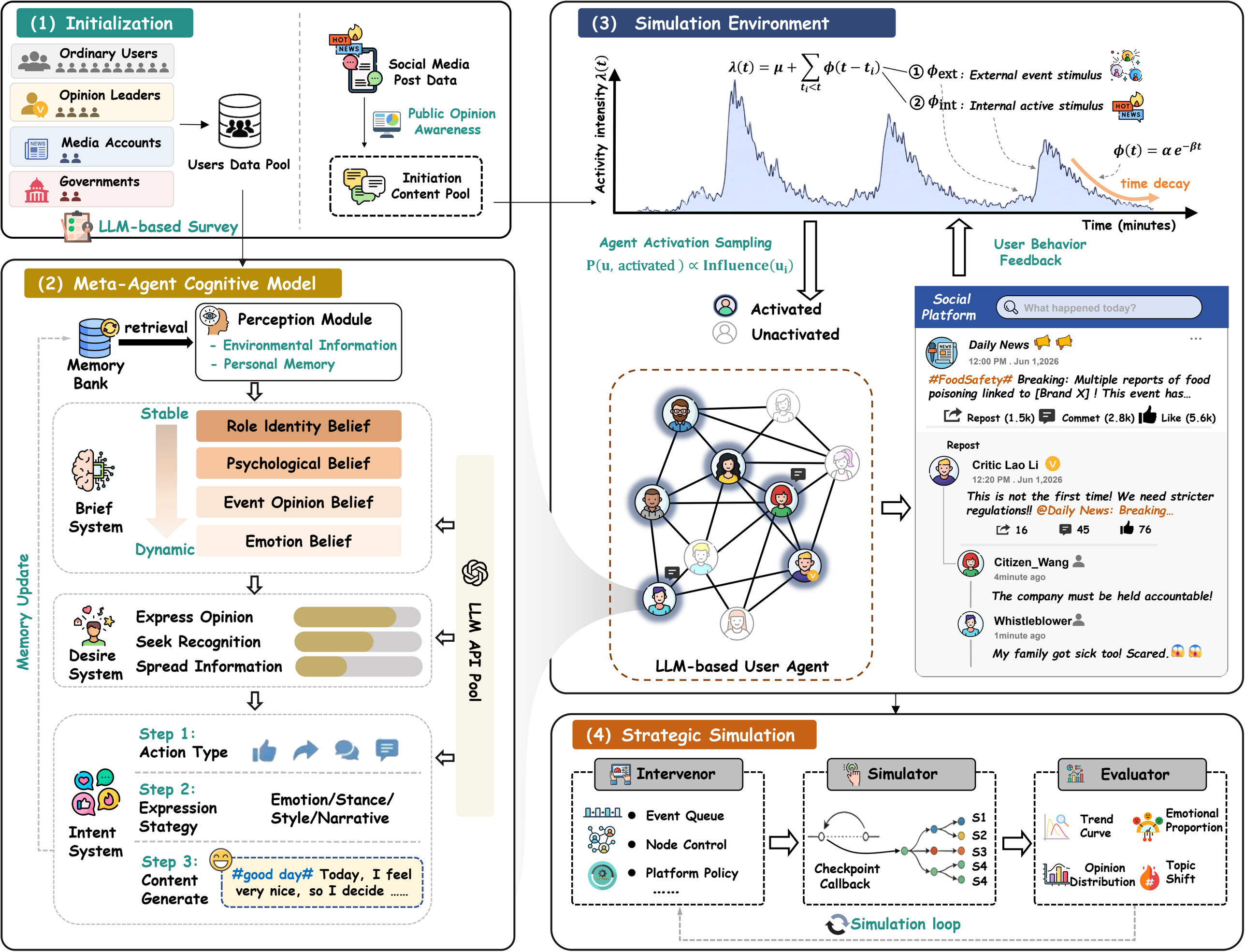
(1) Initialization
Ordinary Users
Opinion Leaders
Media Accounts
Governments
Users Data Pool
LLM-based Survey
Social Media Post Data
Public Opinion Awareness
Initiation Content Pool
(2) Meta-Agent Cognitive Model
retrieval
Memory Bank
Perception Module
- Environmental Information
- Personal Memory
Stable
Dynamic
Brief System
Role Identity Belief
Psychological Belief
Event Opinion Belief
Emotion Belief
Desire System
Express Opinion
Seek Recognition
Spread Information
Intent System
Step 1: Action Type
Step 2: Expression Stategy
Emotion/Stance/ Style/Narrative
Step 3: Content Generate
#good day# Today, I feel very nice, so I decide .....
Memory Update
LLM API Pool
(3) Simulation Environment
$\lambda(t) = \mu + \sum_{t_i<t} \phi(t - t_i)$
① ϕ_{ext} : External event stimulus
② ϕ_{int} : Internal active stimulus
$\phi(t) = \alpha e^{-\beta t}$
time decay
Activity intensity $\lambda(t)$
Time (minutes)
Agent Activation Sampling
$P(u, \text{activated}) \propto \text{Influence}(u_i)$
Activated
Unactivated
User Behavior Feedback
LLM-based User Agent
Social Platform
What happened today?
Daily News
12:00 PM . Jun 1,2026
#FoodSafety# Breaking: Multiple reports of food poisoning linked to [Brand X] ! This event has...
Repost (1.5k)
Commet (2.8k)
Like (5.6k)
Repost
Critic Lao Li
12:20 PM . Jun 1,2026
This is not the first time! We need stricter regulations!! @Daily News: Breaking...
16
45
76
Citizen_Wang
4minute ago
The company must be held accountable!
Whistleblower
1minute ago
My family got sick too! Scared.
(4) Strategic Simulation
Intervenor
Event Queue
Node Control
Platform Policy
.....
Simulator
Checkpoint Callback
S1
S2
S3
S4
S4
Evaluator
Trend Curve
Emotional Proportion
Opinion Distribution
Topic Shift
Simulation loop